

Chapter 6 知识点

6.1 Rotation with Constant Angular Acceleration

Straight-Line Motion with Constant Linear Acceleration	Fixed-Axis Rotation with Constant Angular Acceleration
$a_x = \text{constant}$	$\alpha_z = \text{constant}$
$v_x = v_{0x} + a_x t$	$\omega_z = \omega_{0z} + \alpha_z t$
$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$	$\theta = \theta_0 + \omega_{0z} t + \frac{1}{2} \alpha_z t^2$
$v_x^2 = v_{0x}^2 + 2a_x (x - x_0)$	$\omega_z^2 = \omega_{0z}^2 + 2\alpha_z (\theta - \theta_0)$
$x - x_0 = \frac{1}{2} (v_{0x} + v_x) t$	$\theta - \theta_0 = \frac{1}{2} (\omega_{0z} + \omega_z) t$

Chapter 6 知识点

6.2 The moment of inertia, I ,

不连续的质点系的转动惯量

Moment of inertia
of a body for a given rotation axis

Masses of the particles that make up the body

$$I = m_1 r_1^2 + m_2 r_2^2 + \cdots = \sum_i m_i r_i^2$$

Perpendicular distances of the particles from rotation axis

连续的质点系的转动惯量求解要用积分形式

$$I = \int r^2 dm$$

$$dm = \rho dV \quad \text{or} \quad \delta ds \quad \text{or} \quad \lambda dl$$

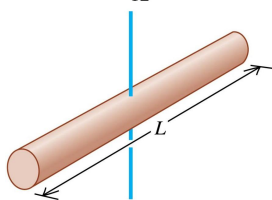
6.3 典型的转动惯量 I

the parallel-axis theorem $I_I = I_{cm} + MR^2$
 I_{cm} 是相对质心的

两个物体的转动惯量可以叠加，但是
 是要相对于同一点。

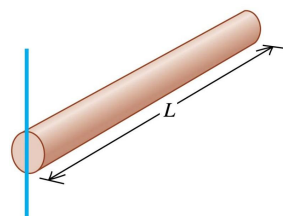
(a) Slender rod,
axis through center

$$I = \frac{1}{12}ML^2$$



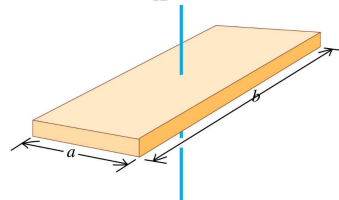
(b) Slender rod,
axis through one end

$$I = \frac{1}{3}ML^2$$



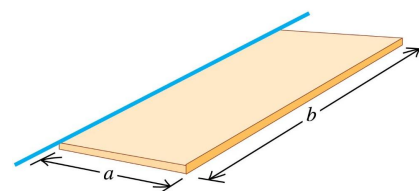
(c) Rectangular plate,
axis through center

$$I = \frac{1}{12}M(a^2 + b^2)$$



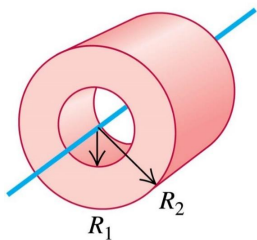
(d) Thin rectangular plate,
axis along edge

$$I = \frac{1}{3}Ma^2$$



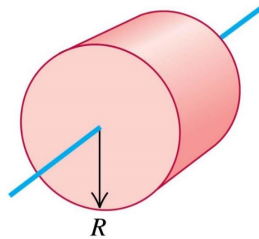
(e) Hollow cylinder

$$I = \frac{1}{2}M(R_1^2 + R_2^2)$$



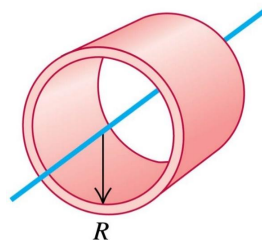
(f) Solid cylinder

$$I = \frac{1}{2}MR^2$$



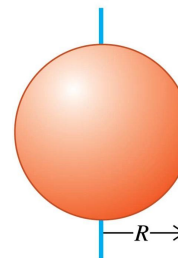
(g) Thin-walled hollow
cylinder

$$I = MR^2$$



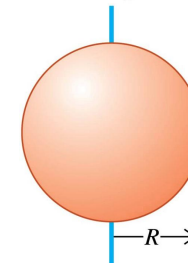
(h) Solid sphere

$$I = \frac{2}{5}MR^2$$



(i) Thin-walled hollow
sphere

$$I = \frac{2}{3}MR^2$$



5-7 匀质大圆盘质量为 M , 半径为 R , 对于过圆心 O 点且垂直于盘面转轴的转动惯量为 $\frac{1}{2}MR^2$. 如果在大圆盘的右半圆上挖去一个小圆盘, 半径为 $R/2$, 如图5-2所示. 剩余部分对于过 O 点且垂直于盘面转轴的转动惯量为_____.

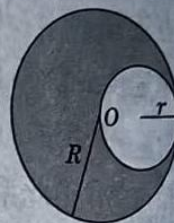


图5-2 习题5-7图

解 运用平行轴定理和转动惯量的可加性, 即可求得 $J = \frac{13}{32}MR^2$.

$$\begin{aligned}
 R &= 2r \\
 I_{O0} &= \frac{1}{2}MR^2 \\
 I_{C0} &= \frac{1}{2}mr^2 \\
 I_{C0} &= I_{C0} + m(\overline{OC})^2 \\
 &= \frac{1}{2}mr^2 + m\left(\frac{R}{2}\right)^2 \\
 &= \frac{3}{2}m\frac{R^2}{4} = \frac{3}{8}mR^2 \\
 I &= I_{O0} - I_{C0} \\
 I &= \frac{1}{2}MR^2 - \frac{3}{8}mR^2 \\
 &= \frac{1}{2}MR^2 - \frac{3}{8} \frac{1}{4}MR^2 = \frac{13}{32}MR^2
 \end{aligned}$$

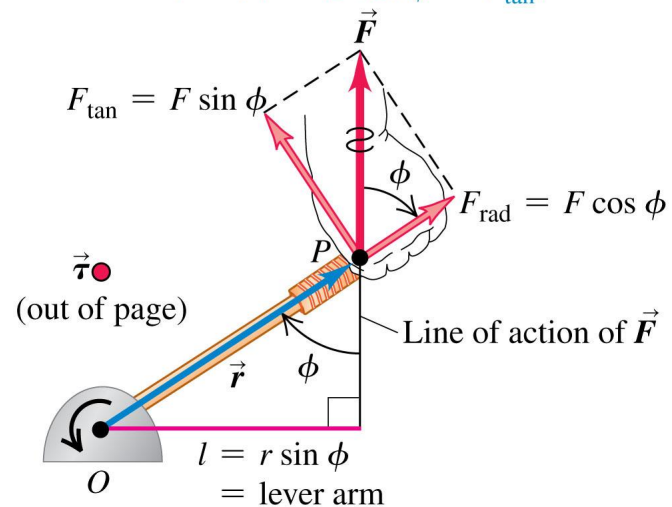
$\left(\frac{m}{M} = \frac{\pi r^2}{\pi R^2} = \frac{1}{4} \right)$

Chapter 7 知识点

7.1 Torque

Three ways to calculate torque:

$$\tau = Fl = rF \sin \phi = F_{\tan} r$$



7.2 The rotational analog of Newton's second law for a rigid body is

Rotational analog of Newton's second law for a rigid body:

Net torque on a rigid body about z -axis

$$\sum \tau_z =$$

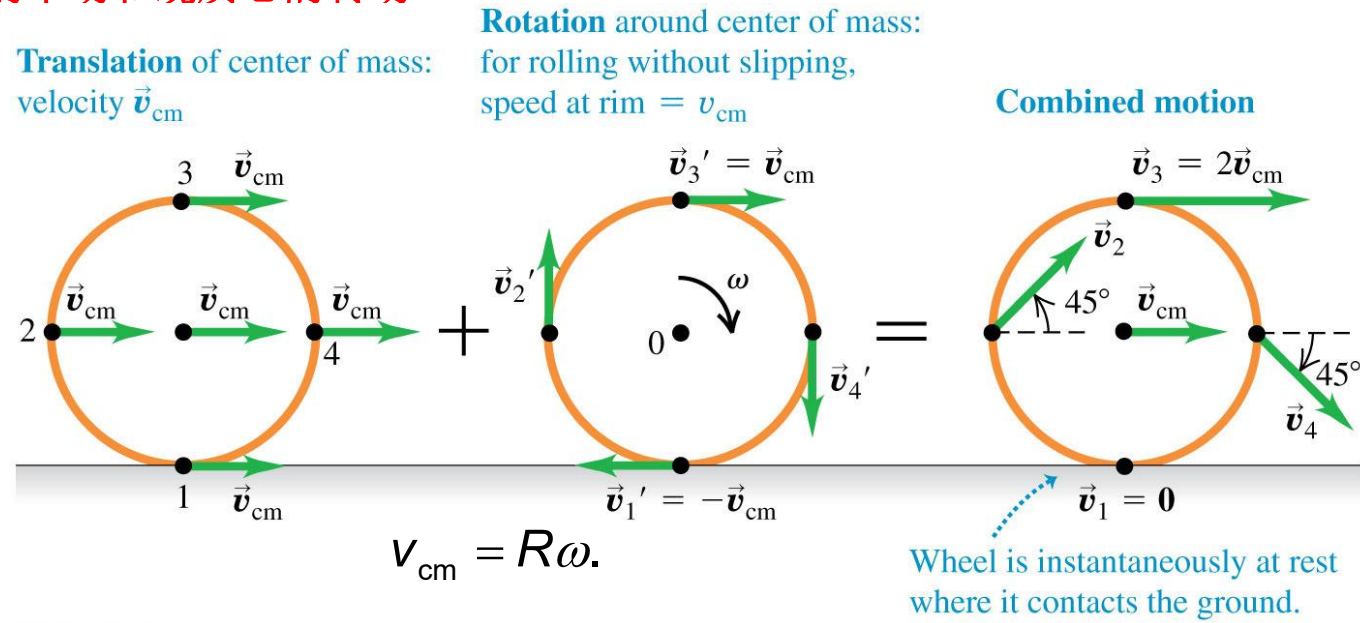
Moment of inertia of rigid body about z -axis

Angular acceleration of rigid body about z -axis

$$I \alpha_z$$

Chapter 7

7.3 Rolling Without Slipping 一边转，一边整体往前动的刚体，转换成质心的平动和绕质心的转动



The kinetic energy of the wheel is

$$K = \frac{1}{2} I_I \omega^2 = \frac{1}{2} I_{\text{cm}} \omega^2 + \frac{1}{2} M R^2 \omega^2 = \frac{1}{2} I_{\text{cm}} \omega^2 + \frac{1}{2} M v_{\text{cm}}^2$$

The gravitational potential energy

$$U = M g y_{\text{cm}}$$

Chapter 7

7.4 Angular Momentum

- For a rigid body rotating around an axis of symmetry the angular momentum is:

Angular momentum of a rigid body rotating around a symmetry axis $\vec{L} = I\vec{\omega}$ Moment of inertia of rigid body about symmetry axis Angular velocity vector of rigid body

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v} \quad (\text{angular momentum of a particle})$$

For any system of particles, the rate of change of the total angular momentum equals the sum of the torques of all forces acting on all the particles:

For a system of particles:

Sum of external torques on the system

$$\sum \vec{\tau} = \frac{d\vec{L}}{dt}$$

Rate of change of total angular momentum $\vec{L}$ of system

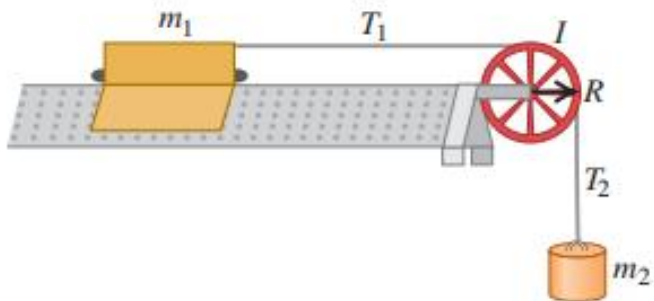
Torque vector due to force $\vec{F}$ relative to point O

$$\vec{\tau} = \vec{r} \times \vec{F}$$

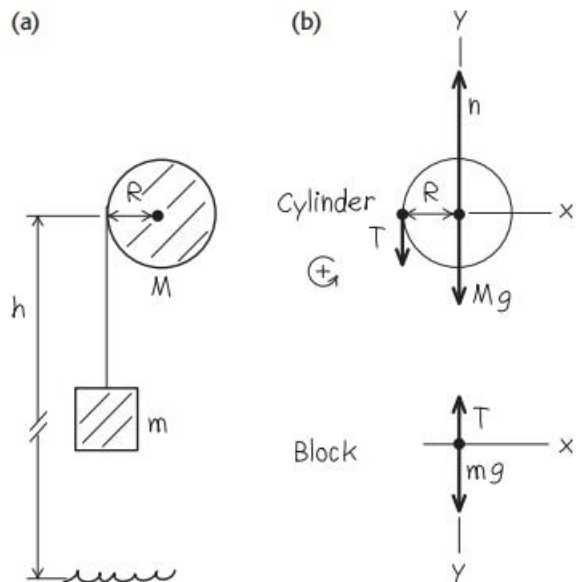
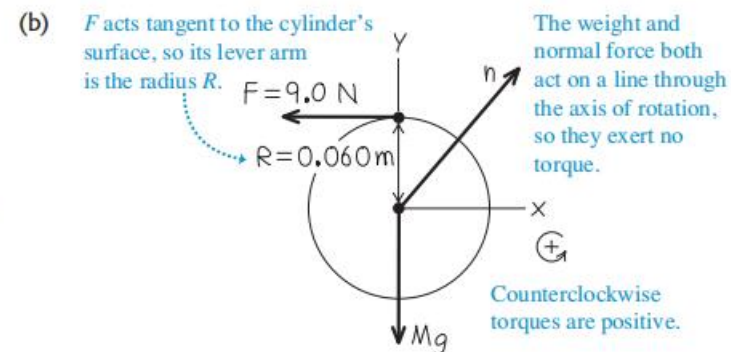
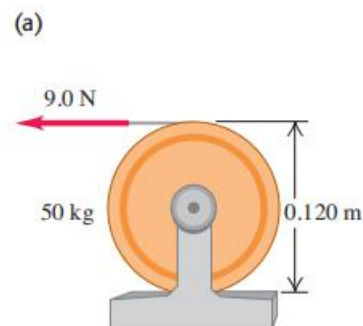
Vector from O to where $\vec{F}$ acts Force $\vec{F}$

	纯平动	纯转动
牛二	$F = ma$	$\tau = I\alpha$
动能	$\frac{1}{2}mv^2$	$\frac{1}{2}I\omega^2$
动量	$\vec{p} = m\vec{v}$	角动量 $\vec{L} = I\vec{\omega}$
动量定理的牛二	$\vec{F} = \frac{d\vec{p}}{dt}$	角动量定理的牛二 $\vec{\tau} = \frac{d\vec{L}}{dt}$
动量守恒条件	合外力 = 合外力 = 0 (内力之和始终为0) $\therefore$ 内力不考虑	角动量守恒条件 合外力矩 = 0 (合内力矩之和为0) $\therefore$ 内力矩不考虑

7.5 滑轮问题



9 (a) Cylinder and cable. (b) Our free-body diagram for the cylinder.



题目中有时间，或者加速度一般的求解方法

1. 找出研究对象
2. 分析受力
3. 列出几个牛二，（刚体用转动的牛二）
4. 没有相对滑动时，转动刚体边缘的的加速度和绳的加速度相等。

7.6 摆动问题和碰撞问题

5-17 如图 5-9 所示, 质量为 m_1 、长为 l 的直杆, 可绕水平轴 O 无摩擦地转动. 设一质量为 m_2 的子弹沿水平方向飞来, 恰好射入杆的下端, 若直杆 (连同射入的子弹) 的最大摆角为 $\theta = 60^\circ$, 试证明子弹的速率为

$$v_0 = \sqrt{\frac{(m_1 + 2m_2)(m_1 + 3m_2)gl}{6m_2^2}}$$

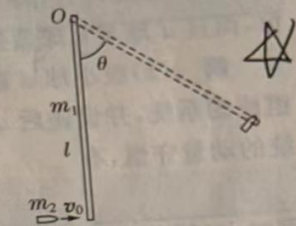


图 5-9 习题 5-17 图

分析 子弹与直杆的运动由两个过程组成. 第一个过程为完全非弹性的碰撞过程, 对于由子弹和直杆构成的系统, 因所受外力对轴的力矩为零, 所以系统对轴的角动量守恒. 第二个过程是子弹与直杆一起上摆的过程, 若仍取子弹和直杆为系统, 在上摆过程中只有重力做功, 利用动能定理即可求得最大摆角. 若取子弹、直杆和地球为系统, 则因系统没有外力做功, 也没有非保守内力做功, 因此系统的机械能守恒.

证明 子弹射入直杆, 根据角动量守恒, 有

$$m_2 v_0 l = \left(\frac{1}{3} m_1 l^2 + m_2 l^2 \right) \omega \quad (1)$$

子弹射入后, 细杆在摆动过程中只有重力做功. 因此, 若以子弹、细杆和地球为一系统, 则此系统机械能守恒, 有

$$\frac{1}{2} \left(\frac{1}{3} m_1 l^2 + m_2 l^2 \right) \omega^2 = m_2 gl (1 - \cos 60^\circ) + m_1 g \frac{l}{2} (1 - \cos 60^\circ)$$

化简, 得

$$2(m_1 + 3m_2)l^2 \omega^2 = 3(m_1 + 2m_2)gl \quad (2)$$

由①、②式, 解得

$$v_0 = \sqrt{\frac{(m_1 + 2m_2)(m_1 + 3m_2)gl}{6m_2^2}}$$

碰撞的一瞬间,
角动量守恒, 动量不守恒 (由于钉子对O有水平力的作业), 接下来只有重力做功机械能守恒

Chapter 8 知识点

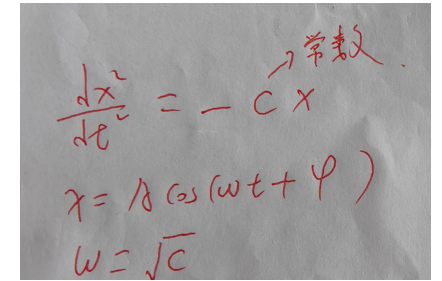
8.1 Simple Harmonic Motion

- The **amplitude**, A , is the maximum magnitude of displacement from equilibrium.
- The **period**, T , is the time for one cycle.
- The **frequency**, f , is the number of cycles per unit time.
- The **angular frequency**, ω , is 2π times the frequency: $\omega = 2\pi f$.
- The frequency and period are reciprocals of each other:

$$T = \frac{1}{f}, f = \frac{1}{T}.$$

证明是否可以进行**SHM**运动的方法

- 1.选坐标系，正反方向没有关系，一定要选受力平衡位置为坐标原点
- 2.用坐标表示，找出任意一点的受力牛二方程。
- 3.如满足未知数的二阶求导等于负数乘以未知数，即可得出方程的解为 $x = A\cos(\omega t + \phi)$ ，是简谐振动。

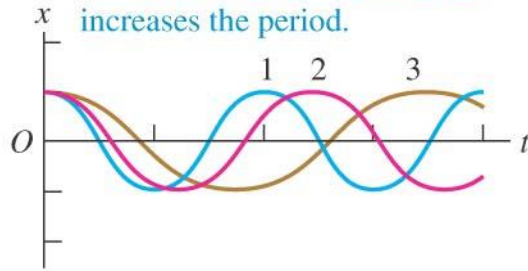


Handwritten notes showing the derivation of SHM equations:

$$\frac{d^2x}{dt^2} = -cx \quad \text{（c 为常数）}$$
$$x = A\cos(\omega t + \phi)$$
$$\omega = \sqrt{c}$$

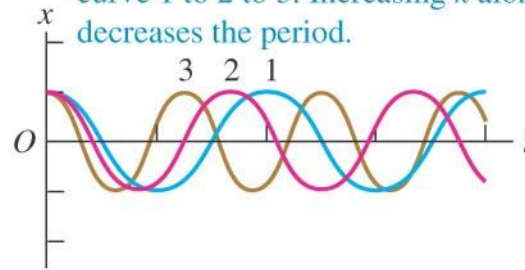
(a) Increasing m ; same A and k

Mass m increases from curve 1 to 2 to 3. Increasing m alone increases the period.



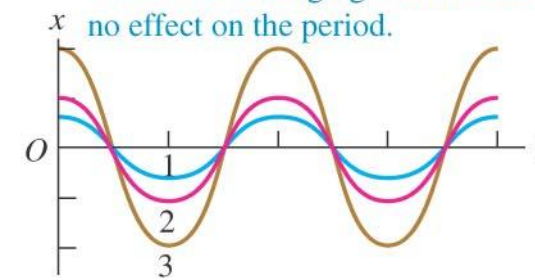
(b) Increasing k ; same A and m

Force constant k increases from curve 1 to 2 to 3. Increasing k alone decreases the period.



(c) Increasing A ; same k and m

Amplitude A increases from curve 1 to 2 to 3. Changing A alone has no effect on the period.



8.2 常见的Simple Harmonic Motion (角频率是本征的量, 不同系统不一样, 下面是三种典型的)

1. 一端固定, 一端挂有物块的弹簧振子, 不管水平还是竖直。

$$\omega = \sqrt{\frac{k}{m}} \quad f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad T = \frac{1}{f} = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$

2. simple pendulum

$$F_{\theta} = -mg\theta = -mg\frac{x}{L},$$

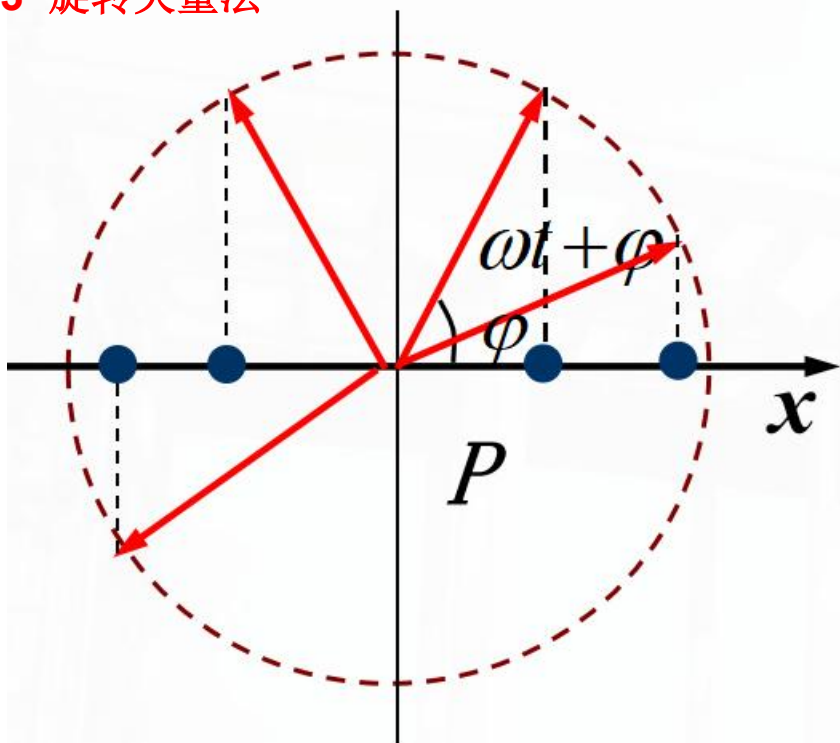
$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{\frac{mg}{L}}{m}} = \sqrt{\frac{g}{L}}, \quad f = \frac{1}{2\pi} \sqrt{\frac{g}{L}}$$

3. physical pendulum

• The angular frequency is

given by
$$\omega = \sqrt{\frac{mgd}{I}}$$

8.3 旋转矢量法



结论： 投影点的运动为简谐运动.

$$x = A \cos(\omega t + \varphi)$$

四、旋转矢量法

- 旋转矢量 $\vec{A}$ 的模即为简谐运动的**振幅**.
- 旋转矢量 $\vec{A}$ 的角速度 ω 即为振动的**角频率**.
- 旋转矢量 $\vec{A}$ 与 x 轴的夹角 $(\omega t + \varphi)$ 为简谐运动的**相位**.
- $t=0$ 时, $\vec{A}$ 与 x 轴的夹角 φ 即为简谐振动的**初相位**.
- 旋转矢量 $\vec{A}$ 旋转一周, P 点完成一次全振动. $T = \frac{2\pi}{\omega}$

例 一质点沿 x 轴作简谐运动，振幅为**12cm**，周期为**2s**。当 **$t = 0$** 时，位移为**6cm**，且向 **x 轴正方向**运动。求：**(1)** 振动表达式；**(2)** **$t = 0.5\text{s}$** 时，质点的位置、速度和加速度；**(3)** 如果在某时刻质点位于 **$x = -0.6\text{cm}$** ，且向 **x 轴负方向**运动，求从该位置回到平衡位置所需要的时间。

解：(1)

$$A = 0.12 \text{ m}$$

$$T = 2 \text{ s}$$

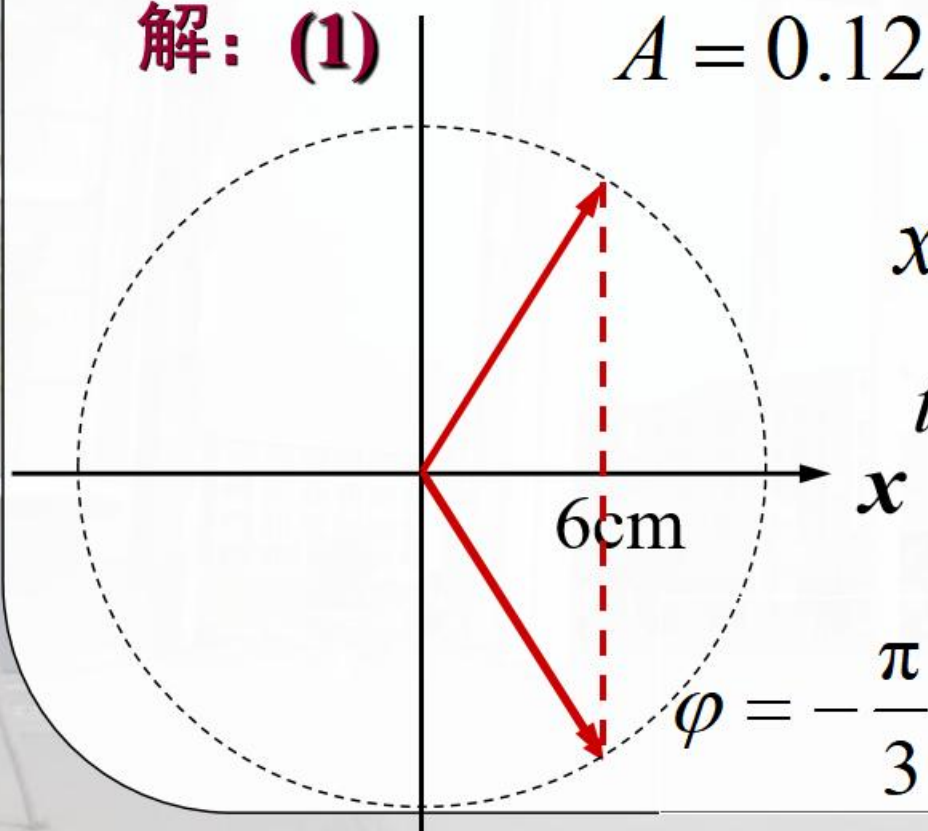
$$\omega = \frac{2\pi}{T} = \pi \text{ s}^{-1}$$

$$x = 0.12 \cos(\pi t + \varphi) \text{ m}$$

$$t = 0 \text{ 时, } x_0 = 0.06 \text{ m, } v_0 > 0$$

振动表达式：

$$x = 0.12 \cos\left(\pi t - \frac{\pi}{3}\right) \text{ m}$$



(2) $t = 0.5\text{s}$ 时, 质点的位置、速度和加速度

$$x\Big|_{t=0.5\text{s}} = 0.12 \cos\left(\pi t - \frac{\pi}{3}\right)\Big|_{t=0.5\text{s}} = -0.10 \text{ m}$$

$$\begin{aligned} v\Big|_{t=0.5\text{s}} &= \frac{dx}{dt}\Big|_{t=0.5\text{s}} = -0.12 \pi \sin\left(\pi t - \frac{\pi}{3}\right)\Big|_{t=0.5\text{s}} \\ &= -0.19 \text{ m/s} \end{aligned}$$

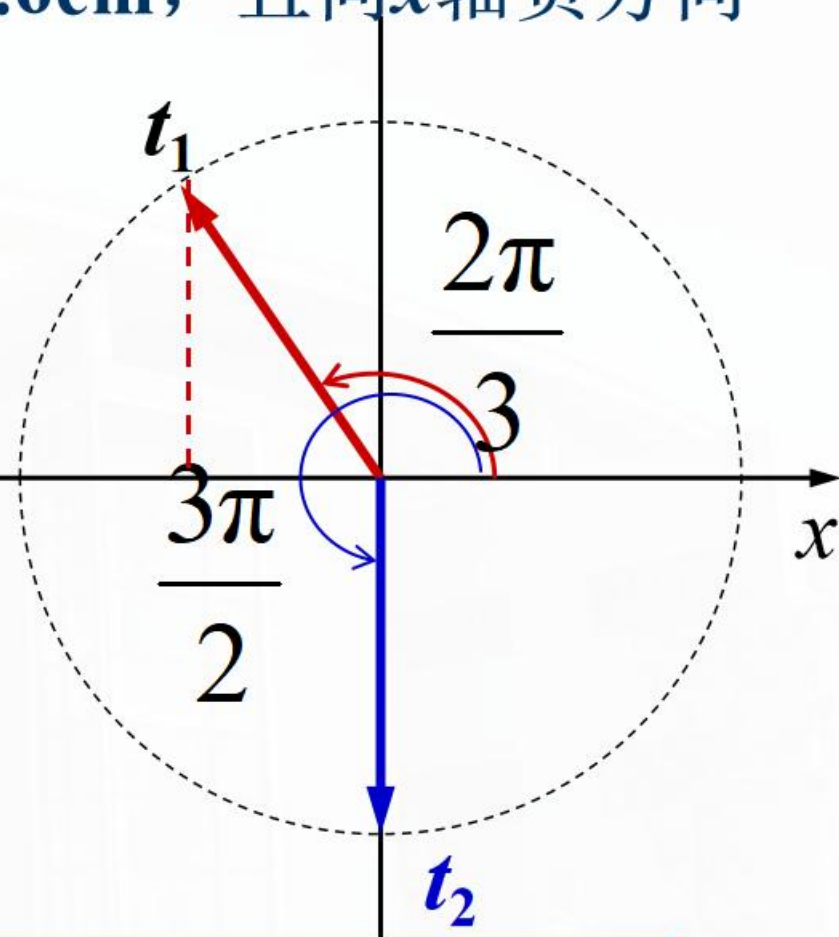
$$\begin{aligned} a\Big|_{t=0.5\text{s}} &= \frac{dv}{dt}\Big|_{t=0.5\text{s}} = -0.12 \pi^2 \cos\left(\pi t - \frac{\pi}{3}\right)\Big|_{t=0.5\text{s}} \\ &= -1.0 \text{ m/s}^2 \end{aligned}$$

(3)如果在某时刻质点位于 $x = -0.6\text{cm}$ ，且向 x 轴负方向运动，求从该位置回到平衡位置所需要的时间。

$$\varphi_1 = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2}{3}\pi \quad \varphi_2 = \frac{3\pi}{2}$$

$$\Delta\varphi = \frac{3\pi}{2} - \frac{2\pi}{3} = \frac{5\pi}{6}$$

$$\Delta\varphi = \omega \Delta t$$



$$\Delta t = \frac{\Delta\varphi}{\omega} = \frac{5\pi}{6\pi} = \frac{5}{6} \text{ s}$$

8.4 简谐运动的合成-同方向同频率简谐运动的合成

某一质点沿 x 轴同时参与两个独立的同频率的简谐运动，其运动表达式分别表示为

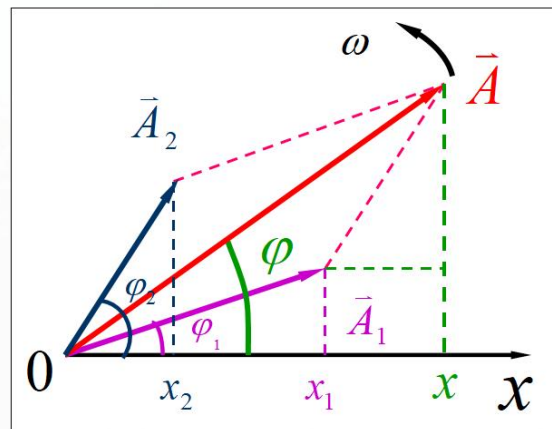
$$x_1 = A_1 \cos(\omega t + \varphi_1) \quad x_2 = A_2 \cos(\omega t + \varphi_2)$$

2. 旋转矢量法

由图可知

$$x = A \cos(\omega t + \varphi)$$

根据平行四边形法则得



$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2(\varphi_2 - \varphi_1)}$$

$$\tan \varphi = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2}$$

仍然是简谐运动

Chapter 9 知识点

9.1 Wave

$$y(x,t) = A\cos(\omega t - kx + \phi) \quad \text{正方向传播}$$

$$y(x,t) = A\cos(\omega t + kx + \phi) \quad \text{负方向传播}$$

波速 v

ω T f

k λ

这些量之间的关系要会
波速和振动速度不一样注意区分

波速是本质的，和介质有关
振动速度是 dy/dt ，不同时刻不一样

$$k = \frac{2\pi}{\lambda} \quad (\text{wave number})$$

$$\lambda = 2\pi/k \quad f = \omega/2\pi \quad v = \lambda f \quad \omega = vk \quad (\text{periodic wave})$$

Chapter 9 知识点

9.2 Principle of Superposition Standing Waves

$$y(x, t) = y_1(x, t) + y_2(x, t) \quad (\text{principle of superposition})$$

At a **node** the displacements of the two waves in red and blue are always equal and opposite and cancel each other out. This cancellation is called **destructive interference**. **波节是相消的结果**

At the **antinodes** the displacements of the two waves in red and blue are always identical, giving a large resultant displacement; this phenomenon is called **constructive interference**. **波腹是增强的结果**

$$y_1 = A \cos(\omega t - kx + \varphi_1)$$

$$y_2 = A \cos(\omega t + kx + \varphi_2)$$

$$y = y_1 + y_2$$

$$= 2A \cos\left(kx + \frac{\varphi_2 - \varphi_1}{2}\right) \cos\left(\omega t + \frac{\varphi_2 + \varphi_1}{2}\right)$$

①

数值法看 $\cos\left(kx + \frac{\varphi_2 - \varphi_1}{2}\right) = 0$ 相消

$$\cos\left(kx + \frac{\varphi_2 - \varphi_1}{2}\right) = \pm 1 \text{ 相增}$$

②

用旋转矢量法看 y_2 的相位 - y_1 的相位

$$= 2kx + \varphi_2 - \varphi_1 = \underbrace{(2n+1)\pi}_{\text{奇数}} \text{ 相消}$$

$n = \pm 1, \pm 2, \dots$

$$2kx + \varphi_2 - \varphi_1 = \underbrace{2n\pi}_{\text{偶数}} \text{ 相增}$$

Chapter 9 知识点

9.3 Normal Modes of a String

$$y(x, t) = y_1(x, t) + y_2(x, t) \quad (\text{principle of superposition})$$

At a **node** the displacements of the two waves in red and blue are always equal and opposite and cancel each other out. This cancellation is called **destructive interference**. **波节是相消的结果**

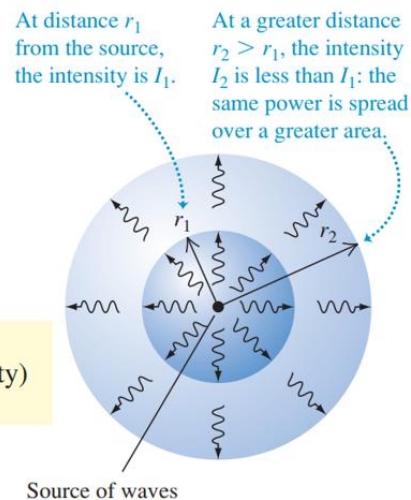
At the **antinodes** the displacements of the two waves in red and blue are always identical, giving a large resultant displacement; this phenomenon is called **constructive interference**. **波腹是增强的结果**

9.4 波能和波强

Wave Intensity

$$4\pi r_1^2 I_1 = 4\pi r_2^2 I_2$$

$$\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2} \quad (\text{inverse-square law for intensity})$$



Energy in Wave Motion

$$\omega = vk \quad v^2 = F/\mu$$

$$P(x, t) = Fk\omega A^2 \sin^2(kx - \omega t)$$



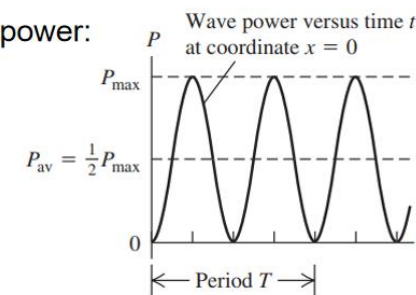
$$P(x, t) = \sqrt{\mu F} \omega^2 A^2 \sin^2(kx - \omega t)$$

The maximum value of the instantaneous power:

$$P_{\max} = \sqrt{\mu F} \omega^2 A^2$$

The average power:

$$P_{\text{av}} = \frac{1}{2} \sqrt{\mu F} \omega^2 A^2$$



Chapter 9 知识点

9.5 多普勒效应

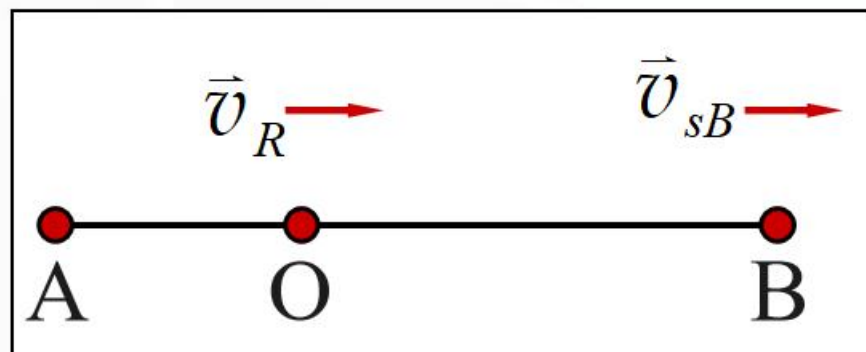
$$f_L = \frac{v \pm v_L}{v \mp v_S} f_S$$

听者，听者靠近
是正，远离是负

源头，源头靠近
是负，远离是正

例1 A、B 为两个汽笛，其频率皆为**50Hz**，A 静止，B 以**60m/s** 的速率向右运动. 在两个汽笛之间有一观察者 O，以**30m/s** 的速度也向右运动. 已知空气中的声速为**330m/s**，求：

- 1) 观察者听到来自A 的频率
- 2) 观察者听到来自B 的频率
- 3) 观察者听到的拍频



解： 1) $u = 330 \text{ m/s}, v_{sA} = 0, v_{sB} = 60 \text{ m/s}$

$$v_R = \frac{330 \text{ m} \cdot \text{s}^{-1} - 30 \text{ m} \cdot \text{s}^{-1}}{330 \text{ m} \cdot \text{s}^{-1}} \times 500 \text{ Hz} = 454.5 \text{ Hz}$$

解： 2)

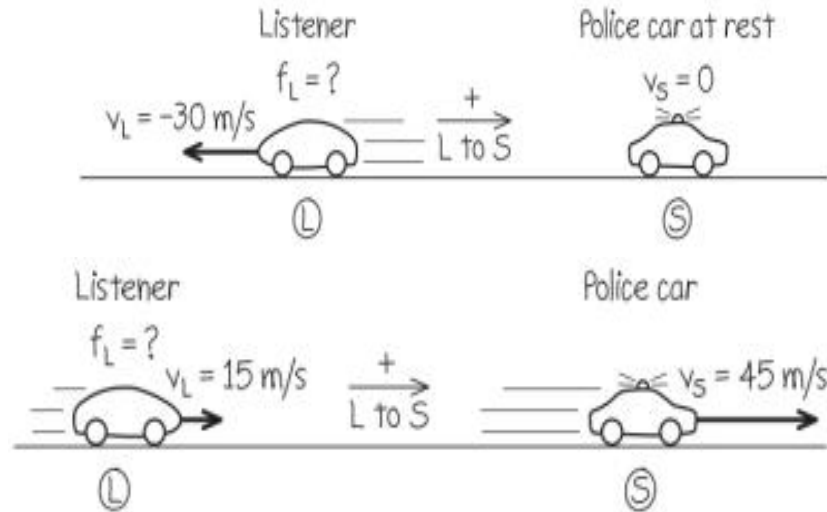
$$\nu'_R = \frac{330\text{m}\cdot\text{s}^{-1} + 30\text{m}\cdot\text{s}^{-1}}{330\text{m}\cdot\text{s}^{-1} + 60\text{m}\cdot\text{s}^{-1}} \times 500\text{Hz} = 461.5\text{Hz}$$

3)

$$\Delta\nu = |\nu' - \nu''| = 7\text{Hz}$$

练习

1. A police car's siren emits a sinusoidal wave with frequency $f_s=150$ Hz. The speed of sound is 340 m/s and the air is still. If the siren is at rest and the listener is moving away from it at 30 m/s. The frequency that listener hear is_____. If the siren is moving away from the listener with a speed of relative to the air, and the listener is moving toward the siren with a speed of relative to the air. The frequency that listener hear is_____.



1. A certain transverse wave is described by $y(x,t)=8\cos(4\pi t+8\pi x+0.5\pi)$ (SI unit). Determine the frequency f , wave length λ , propagation speed v

A particle does simple harmonic oscillation. The period is T , When a particle is moving from its equilibrium position to the positive direction of the x axis, The minimum time required from the equilibrium position to the place of $1/2$ maximum displacement is

6. 一弹簧振子，当把它水平放置时，它可以作简谐振动。若把它竖直放置或放在固定的光滑斜面上，试判断下面哪种情况是正确的：_____。（ ）

- A. 竖直放置可作简谐振动，放在光滑斜面上不能作简谐振动
- B. 竖直放置不能作简谐振动，放在光滑斜面上可作简谐振动
- C. 两种情况都可作简谐振动
- D. 两种情况都不能作简谐振动

